

What is Familiar is Usable: Evaluating the Impact of Familiarity on User Experience and Usability in Mobile App Stores

Christoph Wimmer
Research Group INSO
TU Wien
Vienna, Austria
christoph.wimmer@tuwien.ac.at

Stefan Erceg
Research Group INSO
TU Wien
Vienna, Austria
stefan.erceg@tuwien.ac.at

Thomas Grechenig
Research Group INSO
TU Wien
Vienna, Austria
thomas.grechenig@tuwien.ac.at

Abstract

This study evaluates how familiarity influences UX ratings of Apple App Store and Google Play Store among 40 iPhone and Android users using the User Experience Questionnaire (UEQ). Participants rated their familiar app store highly, while the unfamiliar store scored lower, particularly on pragmatic usability factors like perspicuity, efficiency and dependability. Results highlight the challenge of overcoming resistance to change and the need for alternative platforms to address usability barriers. These findings offer insights for fostering user adoption and competition in mobile app ecosystems.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**.

Keywords

User Experience, App Stores, Familiarity, Change Resistance

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1 Introduction

App stores serve a fundamental role in mobile ecosystems, facilitating the discovery and distribution of applications. Despite the proliferation of alternative app stores, the dominance of established platforms like Apple App Store and Google Play Store remains unchallenged. These platforms have achieved significant market share and established themselves as the primary gateways for app distribution. This entrenched position of dominance has not only shaped the competitive dynamics of the market but also presents significant challenges for emerging players to gain traction. However, the landscape of mobile app stores is undergoing a moment

of transformation. Growing political concern and regulatory pressure surrounding the conduct of dominant platform holders have brought into sharp focus issues of market competition, data privacy, and consumer choice. Regulatory interventions, such as antitrust investigations and legislative proposals, underscore the need for a more equitable and transparent app store ecosystem.

Despite the availability of alternative platforms, users often exhibit a reluctance to deviate from established habits and routines [13]. This resistance manifests in various forms, including inertia, skepticism, and fear of the unknown. While the reasons underlying change resistance are multifaceted, user experience emerges as a critical determinant shaping individuals' attitudes and behaviors towards adopting alternative app stores. User experience [3] encompasses various factors that influence individuals' interactions with mobile app stores, ranging from ease of navigation and search functionality to trust and security considerations. In the context of change resistance, deficiencies in user experience can exacerbate existing barriers to adoption, reinforcing users' reliance on familiar platforms. For instance, cumbersome installation or registration processes, inconsistent app quality, and lack of customer support may deter users from exploring alternative app stores, despite their potential advantages in terms of diversity, innovation, and affordability. Behavioral inertia, driven by a well-established user experience, hinders the adoption of alternative app stores. It is essential to understand and address the UX factors contributing to this resistance to effectively promote competition and innovation in the mobile app distribution market.

This research aims to explore the impact of familiarity (and lack thereof) on various user experience factors using the User Experience Questionnaire (UEQ) [9] in a study involving 40 participants. In doing so, this research not only contributes to our theoretical understanding of user behavior in digital ecosystems but also offers practical implications for designers, policymakers and industry stakeholders seeking to foster a more competitive and inclusive app store landscape.

2 Background

User experience (UX) as defined in ISO 9241-210 is "a person's perceptions and responses that result from the use or anticipated use of a product, system or service" [3]. UX thus deals with all types of responses, from emotional over cognitive to physical, and can be considered as a holistic concept for capturing responses to the actual or anticipated use of a product, system or service [5, 16]. The internal and emotional states of a person using a particular product change before, during and after the use of the product, thus UX is measurable in all three temporal states and can be understood as

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an inherently dynamic process. The goals and needs of a person in relation to contextual environments change over time and have an effect on the perception of UX [17].

UX consists of several factors, which can be measured independently of each other. These include goal-directed or pragmatic usability factors, such as efficiency, controllability and learnability, as well as non-goal-directed or hedonic factors, such as stimulation, novelty, emotions, fun-of-use or aesthetics [16]. In order to measure the UX of a person in relation to a certain product, standardized questionnaires for UX evaluation are suitable, such as AttrakDiff [4], the User Experience Questionnaire (UEQ) [9] or Modular Evaluation of Key Components of User Experience (meCUE) [12] (see section 3.3.1 for more detailed information on questionnaire selection).

Familiar interfaces and procedures provide psychological comfort. Users develop a sense of reliability and predictability with the systems they regularly use, making them less inclined to experiment with unfamiliar systems. These factors contribute to what is often referred to as status quo bias [13], where individuals exhibit a preference for maintaining existing habits and routines over exploring alternatives. Status quo bias has been extensively studied as a cognitive and emotional phenomenon that underpins resistance to change [8, 10]. It is driven by several factors, including loss aversion [6], where individuals perceive potential losses from abandoning a familiar system as more significant than potential gains from adopting a new one.

Prior research highlights how user experience influences resistance to adopting new technologies. Kim and Lee [7] explored how user experience affects resistance to adopting voice user interfaces in in-vehicle infotainment systems. They found that prior experience reduces resistance by lessening transition costs and uncertainty, while moderating the impact of perceived value. Switching costs, particularly transition costs, were key drivers of resistance, highlighting the importance of familiarity and realistic value assessments in mitigating user resistance. Similarly, Dos Santos and Ponchio [14] demonstrated that users with higher prior experience with digital banking services encountered fewer psychological and emotional barriers, such as concerns about risk or image, compared to inexperienced users. Lavorgna et al. [11] explored resistance to digital contact tracing apps during the COVID-19 pandemic, linking sociocultural resistance rooted in doubts about utility, efficiency, and fairness to algorithm aversion and mistrust in technology. Cho et al. [1] investigated factors associated with nurses' resistance to use an electronic health record system and found the relationship between perceived ease of use and resistance to change to be significant. These studies underscore the importance of familiarity in mitigating resistance to technological adoption.

3 Study Design

3.1 Participants

Participants were recruited among existing iPhone and Android smartphone users with a screening questionnaire. Participants were split in two groups according to their smartphone usage: iPhone users, for whom the Apple App Store was familiar and the Google Play Store was unfamiliar, and Android users, for whom the Google Play Store was familiar and the Apple App Store was unfamiliar.

Participants were required (1) to exclusively use either an iPhone or Android smartphone (i.e. no secondary device that belongs to the opposite category), (2) to have used their platform's app store in the past three months, and (3) to not have used the opposite platform's app store in the past 12 months (e.g. after switching platforms) in order to ensure participant's limited exposure to the unfamiliar app store ecosystem. In total, 40 voluntary participants (30 male, 10 female; ages 18-27: 55%; 28-37: 30%; 38-52: 7.5%; 53+: 7.5%) were recruited for the study, with 21 participants being iPhone users and 19 Android users. Participants received no compensation for their participation in the study.

3.2 Procedure

Participants attended on-site sessions, starting with instructions for the session and signing a consent form. The test sessions involved the following key steps in chronological order:

- (1) UEQ for familiar app store: Participants first completed the UEQ for their familiar app store based on prior experience.
- (2) Semi-structured interviews: Interviews were conducted to gather qualitative insights on their experiences with the familiar app store.
- (3) Task performance: Participants performed nine predefined tasks in the unfamiliar app store in order to familiarize themselves.
- (4) UEQ for unfamiliar app store: Participants completed the UEQ again, this time for the unfamiliar app store.
- (5) Follow-up interviews: Additional semi-structured interviews focused on the participants' experiences with the unfamiliar app store.

The semi-structured interviews (steps 2 and 5) consisted of the following questions for both familiar and unfamiliar app stores:

- Are you generally positively or negatively inclined towards the app store?
- Which reasons are crucial for you to have a positive or negative attitude towards the app store?
- Which aspects do you particularly like about the app store and should definitely be kept?
- Which aspects do you think should be improved in the app store?

In addition, the following questions were added in the follow-up interview for unfamiliar app stores (step 5):

- Would you accept that the app store you have just tested replaces the one you use regularly on your personal device? Please explain your opinion.
- Which support methods do you think the development team of an app store could consider and apply in order to minimise the resistance by people to use the store?

To gain an impression of the unfamiliar app store, participants were asked to perform the following nine tasks (step 3):

- (1) Install the app which is ranked in 9th place of the top free apps.
- (2) Sort the reviews of the "Google Maps" app by newest first.
- (3) Open the developer website of the "Zalando" app that is specified in the store.

Table 1: Task success rates and mean completion times of Apple App Store by Android users and Google Play Store by iOS users. Mean completion times based on successful task completions only.

Task	Apple App Store			Google Play Store		
	Success N	Success rate	Completion time	Success N	Success rate	Completion time
Task 1	14	73.68%	00:45	21	100.00%	00:32
Task 2	19	100.00%	00:43	21	100.00%	00:49
Task 3	18	94.74%	01:07	14	66.67%	01:22
Task 4	18	94.74%	00:59	21	100.00%	00:24
Task 5	11	57.89%	01:22	20	95.24%	00:54
Task 6	18	94.74%	00:20	18	85.71%	00:36
Task 7	11	57.89%	01:09	20	95.24%	00:24
Task 8	16	84.21%	00:39	16	76.19%	00:58
Task 9	18	94.74%	00:15	21	100.00%	00:21
Total		83.63%	00:48		91.01	00:42

- (4) Find similar apps to "Uber" (not necessarily from the same developer) and install one of them.
- (5) Check the #1 top free and paid apps in the "News" category.
- (6) Check for pending updates and update apps with available updates.
- (7) Check which subscriptions are currently active.
- (8) Open the overview of all self-written reviews.
- (9) Redeem a voucher code.

A smartphone, either an iPhone for Android users or a Samsung Android smartphone for iOS users, were provided to participants to perform the tasks (step 3). Following the session, participants were debriefed and thanked for their participation.

3.3 Data Collection

UEQ questionnaires were the primary data source in this study and provided quantitative data for statistical analysis. In addition, task performance metrics, including time taken and success rates, were recorded and analyzed. Semi-structured interviews provided qualitative data for analysis. Interviews (see section 3.2, steps 2 and 5) were audio recorded and task performance (see section 3.2, step 3) was screen recorded.

3.3.1 Questionnaire Selection. Prior research by Díaz-Oreiro et al. [2] analyzed which standardized questionnaires were used to evaluate user experience (UX) in previous studies. They reviewed 553 studies and found that 341 studies (61.6%) used AttrakDiff, 200 studies (36.2%) used the User Experience Questionnaire (UEQ), and 12 studies (2.2%) used the modular evaluation of key Components of User Experience (meCUE). Although AttrakDiff was the most frequently used questionnaire initially, the UEQ saw increased usage in studies from 2017 onwards. Additionally, in 340 of the 553 studies (61.5%), another evaluation method was used alongside the standardized questionnaire to assess UX, while 213 studies (38.5%) used only the standardized questionnaire. Specifically, 120 of the 340 studies (35.3%) used the System Usability Scale (SUS) questionnaire to evaluate usability, and 60 of the 340 studies (17.6%) conducted semi-structured interviews.

The decision to use the UEQ in this study was made to comprehensively capture the UX of the systems used by participants.

Unlike AttrakDiff, the UEQ provides a more granular division of pragmatic and hedonic quality factors, allowing for a detailed analysis of the impact of these individual factors on resistance to change. The additional attractiveness factor in the UEQ facilitates both a comprehensive analysis of all factors and the possibility of a standalone assessment. The widespread use of the UEQ in other studies further supported this choice; as Díaz-Oreiro et al. [2] highlighted, the UEQ was employed in 200 out of 553 studies, compared to just 12 using the meCUE. The SUS was not considered for this study because it focuses solely on usability, whereas the UEQ encompasses a broader range of UX aspects, including usability. This comprehensive coverage aligns better with the study's objectives, which aimed to evaluate various dimensions of UX, not just usability.

3.3.2 User Experience Questionnaire (UEQ). The User Experience Questionnaire (UEQ) defines six UX factors that are covered by the questionnaire. The factors are perspicuity, efficiency, dependability, stimulation, novelty and attractiveness. While perspicuity, efficiency and dependability are considered as pragmatic quality factors, which are goal-directed or task-related, stimulation and novelty are understood as hedonic quality aspects and are non-goal-directed or non-task-related. Attractiveness is a pure valence dimension and can be understood as a composition of the other five factors, but can also stand alone [16]. The UEQ consists of 26 items, whereby each item is rated on a seven-point scale and is characterised by a bipolar adjective pair, such as "boring - exciting" [9].

4 Results

4.1 Task Performance

Participants were instructed to perform nine tasks, iPhone users on Google Play Store and Android users on Apple App Store, in order to familiarize themselves with the unfamiliar app store (see section 3.2). Task success rates and completion times were recorded to gain insights into which app store is easier to learn for novice use and to assess the impact of task performance results on subsequent UEQ ratings.

Task performance results (see Table 1) show that overall iOS users were more successful in completing tasks on Google Play

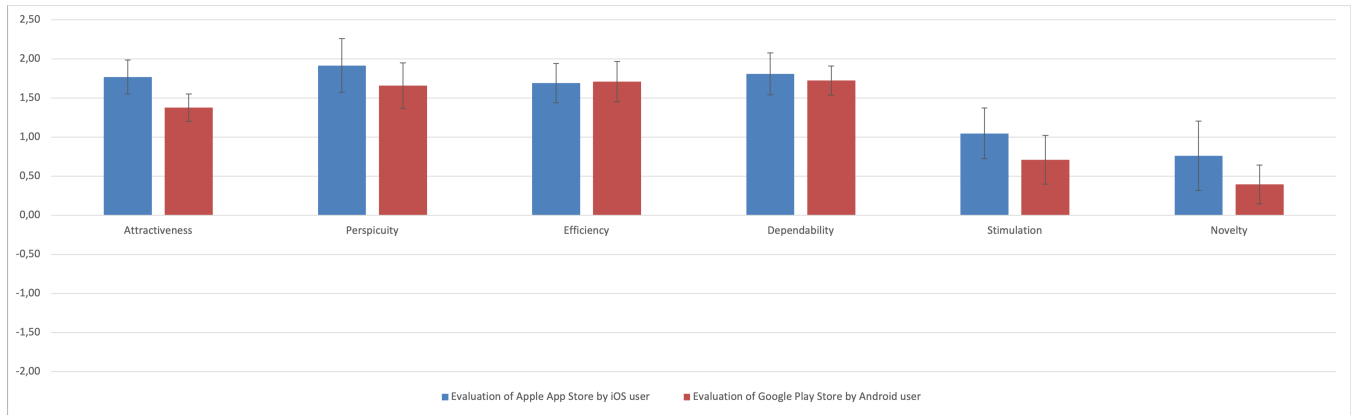


Figure 1: Comparison of UEQ ratings of Apple App Store by iOS users and Google Play Store by Android users categorised by UX factors

Store (success rate 91.01%) compared to Android users on Apple App Store (success rate 83.63%). Task 3 ("Open the developer website of the "Zalando" app that is specified in the store", success rate 66.67%) and task 8 ("Open the overview of all self-written reviews", success rate 76.19%) were the most challenging tasks for iOS users on Google Play Store, while task 5 ("Check the #1 top free and paid apps in the "News" category", success rate 57.89%) and task 7 ("Check which subscriptions are currently active", success rate 57.89%) were the most challenging tasks for Android users on Apple App Store.

4.2 UEQ Rating Results

A repeated measures ANOVA with familiarity (familiar, unfamiliar) as within subject factor and platform user group (Android, iOS) as between subject factor found a significant main effect of familiarity on attractiveness ($F [1, 38] = 73.165, p < 0.001, \eta_p^2 = 0.658$), perspicuity ($F [1, 38] = 411.7, p < 0.001, \eta_p^2 = 0.916$), efficiency ($F [1, 38] = 217.10, p < 0.001, \eta_p^2 = 0.851$), dependability ($F [1, 38] = 393.47, p < 0.001, \eta_p^2 = 0.912$), stimulation ($F [1, 38] = 14.260, p < 0.001, \eta_p^2 = 0.273$) and novelty ($F [1, 38] = 17.76, p < 0.001, \eta_p^2 = 0.319$), and a significant main effect of platform user group on attractiveness ($F [1, 38] = 13.8, p < 0.001, \eta_p^2 = 0.266$), perspicuity ($F [1, 38] = 26.4, p < 0.001, \eta_p^2 = 0.410$), efficiency ($F [1, 38] = 5.18, p = 0.029, \eta_p^2 = 0.120$) and dependability ($F [1, 38] = 10.1, p = 0.003, \eta_p^2 = 0.210$), but not stimulation or novelty. There was a statistically significant interaction effect between familiarity and group on perspicuity ($F [1, 38] = 11.6, p = 0.002, \eta_p^2 = 0.234$) and dependability ($F [1, 38] = 4.54, p = 0.040, \eta_p^2 = 0.107$), but not attractiveness, efficiency, stimulation or novelty.

To establish baseline UEQ ratings for both Apple App Store and Google Play Store, the UEQ ratings of iPhone users rating the App Store and Android users rating the Play Store were analyzed (see Table 3, line "AF - iF"). Figure 1 shows a comparison of the two groups visualizing the arithmetic mean and confidence interval per subordinate UX factor and group. Post-hoc tests show that attractiveness is the only UX factor with a significantly different

rating ($p = 0.045$) for this comparison. Despite small differences in the baseline ratings of Apple App Store and Google Play Store, results for both suggest that their users are overall largely satisfied with the default app stores on their mobile platform of choice. The User Experience Questionnaire Handbook [15] suggests that any rating below -0.8 is negative, ratings between -0.8 and 0.8 are neutral, and ratings above 0.8 are positive. As such, the results show that iPhone users rated five out of six UX factors positively and one neutral, whereas Android users rated four out of six UX factors positively and two neutral. Both iPhone and Android users rated the novelty factor lowest, which can be explained by the fact that they were specifically asked to evaluate a familiar (rather than a novel) experience.

Post-hoc tests show a significant effect of familiarity on attractiveness, perspicuity, efficiency and dependability for all comparisons both within (see Table 3, lines "AF - AU", Android users rating Play Store and App Store, and "iF - iU", iOS users rating App Store and Play Store) and between (see Table 3, lines "AF - iU", both platform user groups rating the Play Store, and "iF - AU", both platform user groups rating the App Store) platform user groups. In addition, there was a significant effect of familiarity on stimulation for comparisons "iF - AU" and "iF - iU" and on novelty for comparisons "AF - iU", "iF - AU" and "iF - iU".

For the AF - AU comparison, the mean difference in ratings is largest for pragmatic UX factors perspicuity ($MD = 3.461$), dependability ($MD = 2.895$) and efficiency ($MD = 2.711$). For the iF - iU comparison, the mean difference in ratings is once again largest for pragmatic UX factors perspicuity ($MD = 2.464$), dependability ($MD = 2.333$) and efficiency ($MD = 2.071$). For the AF - iU comparison, the mean difference in ratings is largest for pragmatic UX factors dependability ($MD = 2.248$), perspicuity ($MD = 2.206$) and efficiency ($MD = 2.092$). For the iF - AU comparison, the mean difference in ratings is once again largest for pragmatic UX factors perspicuity ($MD = 3.719$), dependability ($MD = 2.981$) and efficiency ($MD = 2.691$).

Table 2: Repeated Measures ANOVA

	Attractiveness			Perspicuity			Efficiency		
	F	p	η_p^2	F	p	η_p^2	F	p	η_p^2
Familiarity	73.165	< 0.001	0.658	411.7	< 0.001	0.916	217.10	< 0.001	0.851
Group	13.8	< 0.001	0.266	26.4	< 0.001	0.410	5.18	0.029	0.120
Familiarity $\times$ Group	0.721	0.401	0.019	11.6	0.002	0.234	3.88	0.056	0.093
	Dependability			Stimulation			Novelty		
	F	p	η_p^2	F	p	η_p^2	F	p	η_p^2
Familiarity	393.47	< 0.001	0.912	14.260	< 0.001	0.273	17.76	< 0.001	0.319
Group	10.1	0.003	0.210	1.23	0.275	0.031	1.91	0.175	0.048
Familiarity $\times$ Group	4.54	0.040	0.107	0.275	0.603	0.007	3.66	0.063	0.088

Table 3: Post Hoc Comparisons - Familiarity $\times$ Group (Abbr.: A - Android, i - iOS, F - familiar, U - unfamiliar)

Comparison	Attractiveness				Perspicuity				Efficiency			
	MD	SE	t	<i>P</i> _{tuke}	MD	SE	t	<i>P</i> _{tuke}	MD	SE	t	<i>P</i> _{tuke}
AF - iF	-0.393	0.144	-2.73	0.045	-0.259	0.232	-1.11	0.683	0.020	0.184	0.11	1.000
- AU	1.860	0.287	6.49	< 0.001	3.461	0.212	16.36	< 0.001	2.711	0.235	11.53	< 0.001
- iU	1.131	0.245	4.62	< 0.001	2.206	0.209	10.57	< 0.001	2.092	0.208	10.07	< 0.001
iF - AU	2.252	0.253	8.90	< 0.001	3.719	0.206	18.05	< 0.001	2.691	0.210	12.80	< 0.001
- iU	1.524	0.273	5.59	< 0.001	2.464	0.201	12.25	< 0.001	2.071	0.224	9.26	< 0.001
AU - iU	-0.728	0.321	-2.27	0.124	-1.255	0.179	-7.01	< 0.001	-0.619	0.232	-2.67	0.052
Comparison	Dependability				Stimulation				Novelty			
	MD	SE	t	<i>P</i> _{tuke}	MD	SE	t	<i>P</i> _{tuke}	MD	SE	t	<i>P</i> _{tuke}
AF - iF	-0.086	0.170	-0.50	0.958	-0.337	0.231	-1.46	0.472	-0.367	0.267	-1.38	0.522
- AU	2.895	0.191	15.16	< 0.001	0.855	0.381	2.24	0.130	0.684	0.431	1.59	0.397
- iU	2.248	0.175	12.86	< 0.001	0.794	0.315	2.52	0.072	1.454	0.328	4.43	< 0.001
iF - AU	2.981	0.175	17.01	< 0.001	1.192	0.322	3.70	0.004	1.051	0.334	3.15	0.016
- iU	2.333	0.182	12.85	< 0.001	1.131	0.363	3.12	0.017	1.821	0.410	4.44	< 0.001
AU - iU	-0.647	0.180	-3.60	0.005	-0.061	0.387	-0.159	0.999	0.770	0.385	2.00	0.206

5 Discussion

Baseline ratings indicate that Apple App Store is marginally better rated by iOS users than Google Play Store by Android users. However, these differences are not statistically significant for five out of six factors. Overall, both platforms are seemingly deemed satisfactory by existing users who are familiar with their respective app stores.

Looking at the rating differences between platform user groups, iOS users rated their unfamiliar App Store experience significantly worse for all six UX factors, while Android users rated their unfamiliar App Store experience significantly worse for four out of six UX factors. Comparing the rating differences by app store, Apple App Store was rated significantly worse by Android users compared to iOS users on all six UX factors, and Google Play Store was rated significantly worse by iOS users compared to Android users on five out of six UX factors.

Rating differences were most pronounced for pragmatic UX factors perspicuity, efficiency and dependability for both Apple App Store and Google Play Store, suggesting that these pragmatic factors

have the largest impact on perceived user experience and by extension warrant particular attention by those intending to develop and deploy alternative app stores.

Lower task success rates by Android users on the Apple App Store compared to iOS users on Google Play Store are reflected in particularly low UEQ ratings for pragmatic UX factors. Both lower task success rates and lower UEQ ratings for pragmatic UX factors suggest that the Apple App Store is more difficult to learn than Google Play Store.

Rating differences were more pronounced for Apple App Store than Google Play Store for all UX factors except novelty. We suspect this could be explained by more extensive prior exposure of iOS users to other Google apps and services, resulting in greater inadvertent familiarity with Google design idioms among iOS users, resulting in better task performance, higher ratings for attractiveness and pragmatic UX factors, but also a diminished sense of novelty among iOS users regarding Google Play Store. This assumption was also supported by qualitative feedback from participants.

The results demonstrate a profound and statistically significant impact of familiarity on perceived user experience, with differences

most pronounced for pragmatic usability factors. While such an effect of familiarity might seem obvious and predictable in hindsight, the magnitude of this effect across two different user groups and two different systems under investigation came as a surprise to us. Familiarity as a barrier to competition underscores the limitations of regulatory interventions that focus solely on legal or economic measures. Regulatory frameworks should consider initiatives to promote user education and awareness about alternative platforms. Policymakers might also explore mandating interoperability features, such as seamless app transfers between platforms, to lower switching costs. The results emphasize the need for alternative app stores to minimize the learning curve for new users. Design strategies such as incorporating interface elements similar to dominant platforms or providing comprehensive onboarding experiences could mitigate user resistance. For example, alternative app stores might adopt familiar navigation patterns, terminology, or visual layouts to ease the transition for new users.

Several limitations of this study warrant consideration. The relatively small sample size may limit the generalizability of the findings. Additionally, this study focused on short-term interactions with unfamiliar platforms. Longitudinal studies exploring sustained exposure could provide a deeper understanding of familiarity's influence on UX perceptions. Further research could also investigate how specific UX changes impact the willingness to adopt alternative app stores.

6 Conclusion

Overall, this study highlights the critical role of familiarity in shaping user experience and the barriers it poses to the adoption of alternative app stores. Addressing these barriers requires a multifaceted approach that combines user-centered design, behavioral insights, and supportive regulatory frameworks. Designers of alternative platforms should prioritize pragmatic UX factors while leveraging novelty and innovation to create compelling value propositions. Policymakers and industry stakeholders must work collaboratively to foster a competitive and inclusive mobile app ecosystem that empowers users to make informed choices.

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